(c) e.g. sphere not smooth, humid air, etc ....B1 [1]

5 (a) centripetal force = mv^2/r B1
magnetic force $F = Bqv$ B1
(hence) $mv^2/r = Bqv$ B1
$r = mv/Bq$ ....A0 [3]

(b) $r_\alpha/r_\beta = (m_\alpha/m_\beta) \times (q_\beta/q_\alpha)$ C1
$= (4 \times 1.66 \times 10^{-27})/(9.11 \times 10^{-31} \times 2)$
$= 3.64 \times 10^3$ ....A2 [3]

(c) (i) $r_\alpha = (4 \times 1.66 \times 10^{-27} \times 1.5 \times 10^6)/(1.2 \times 10^{-3} \times 2 \times 1.6 \times 10^{-19})$
$= 25.9$ m A2

(ii) $r_\beta = 25.9 \times 3.64 \times 10^3 = 7.13 \times 10^{-3}$ m ....A1 [3]

(d) (i) deflected upwards..B1
but close to original direction B1

(ii) opposite direction to α-particle and 'through side' ....B1 [3]

6 (a) greater binding energy gives rise to release of energy.................. M1
so must be yttrium ....A1 [2]